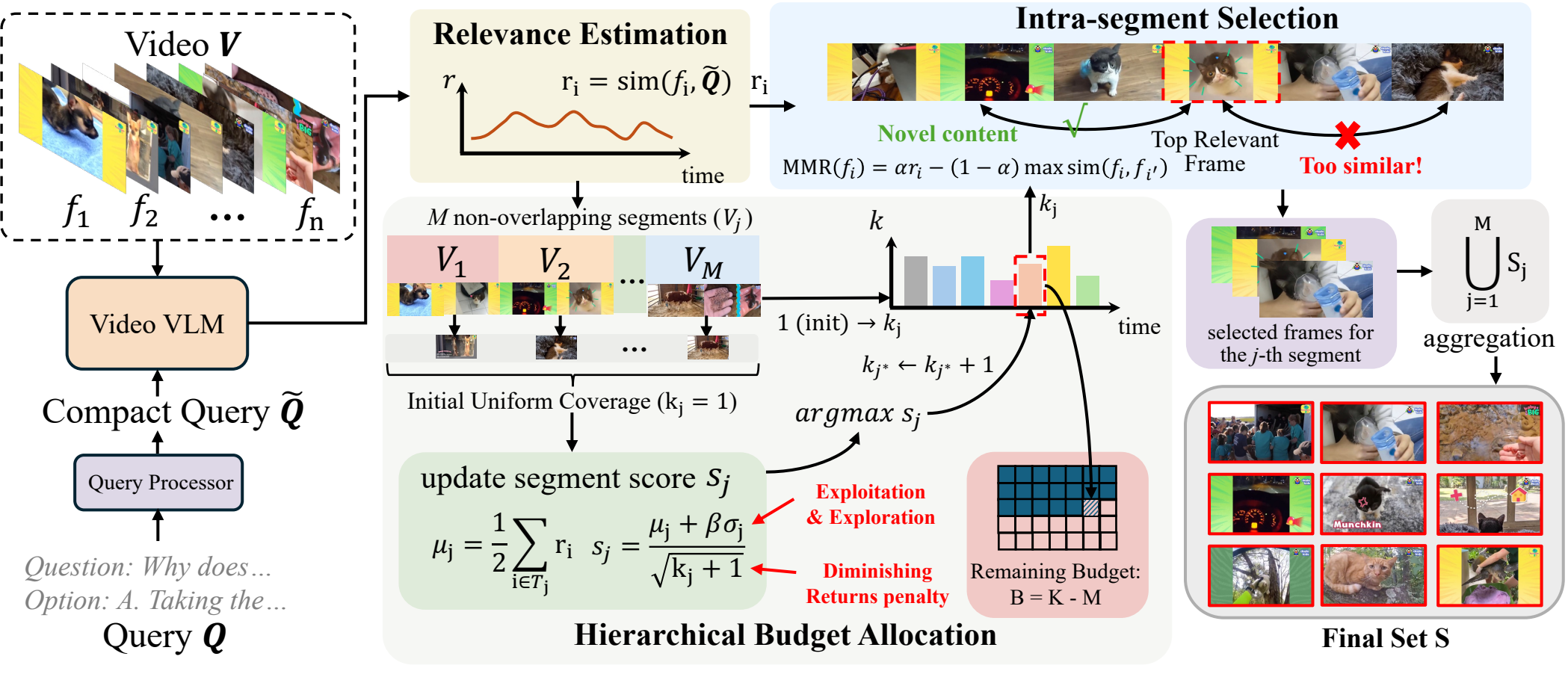
Video V
f_1 f_2 ... f_n
Video VLM
Compact Query $\tilde{Q}$
Query Processor
Question: Why does…
Option: A. Taking the…
Query Q
Relevance Estimation
$r_i = \mathrm{sim}(f_i, \tilde{Q})$
r
time
r_i
Intra-segment Selection
Novel content
Top Relevant Frame
Too similar!
$\mathrm{MMR}(f_i) = \alpha r_i - (1-\alpha)\max \mathrm{sim}(f_i, f_{i'})$
M non-overlapping segments (V_j)
V_1 V_2 ... V_M
Initial Uniform Coverage ($k_j = 1$)
k
k_j
time
1 (init) $\rightarrow k_j$
$k_{j^*} \leftarrow k_{j^*} + 1$
$argmax\ s_j$
update segment score s_j
$\mu_j = \frac{1}{2}\sum_{i \in T_j} r_i \quad s_j = \frac{\mu_j + \beta\sigma_j}{\sqrt{k_j + 1}}$
Exploitation & Exploration
Diminishing Returns penalty
Remaining Budget: B = K - M
Hierarchical Budget Allocation
selected frames for the j-th segment
$\bigcup_{j=1}^{M} S_j$
aggregation
Final Set S